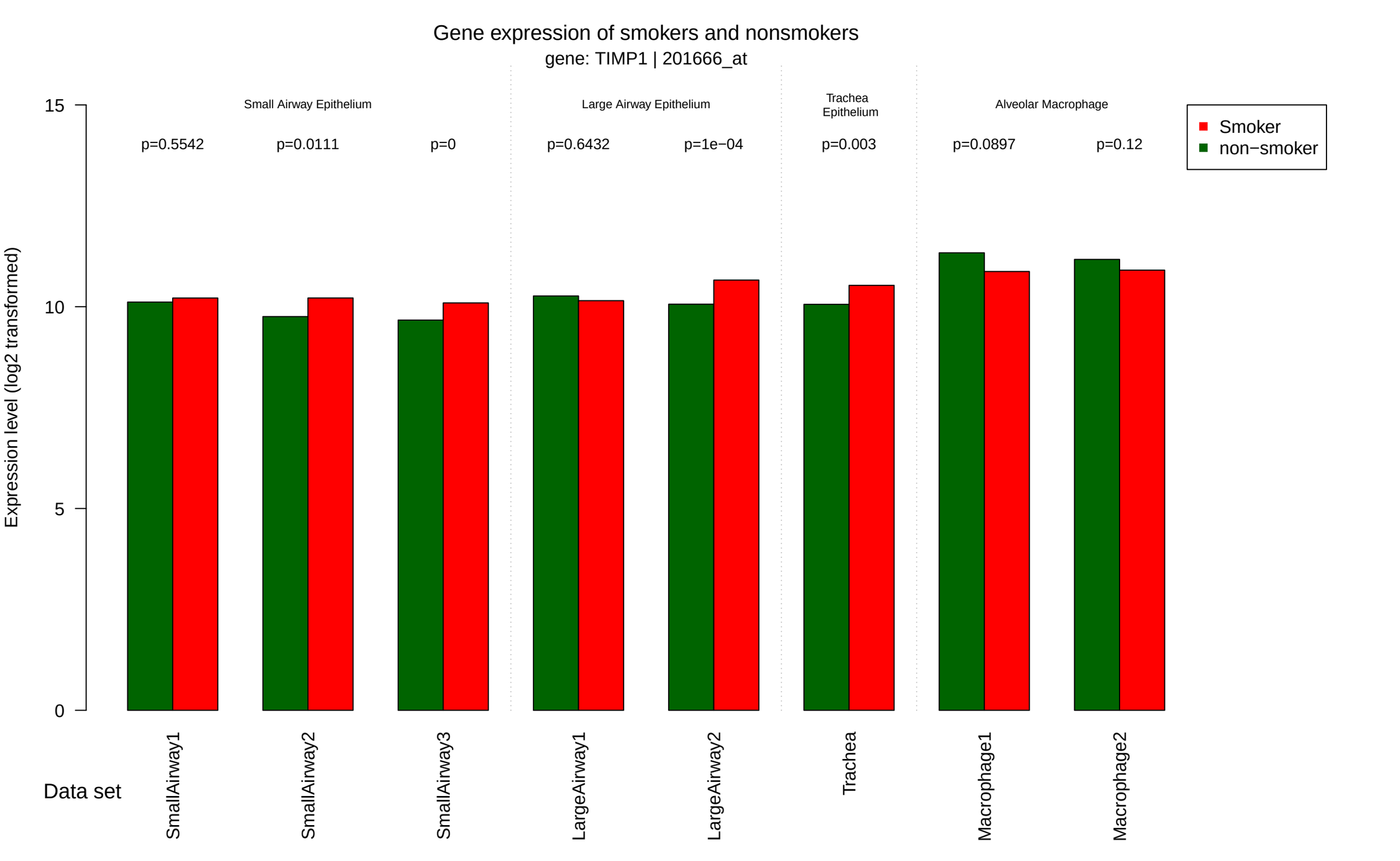
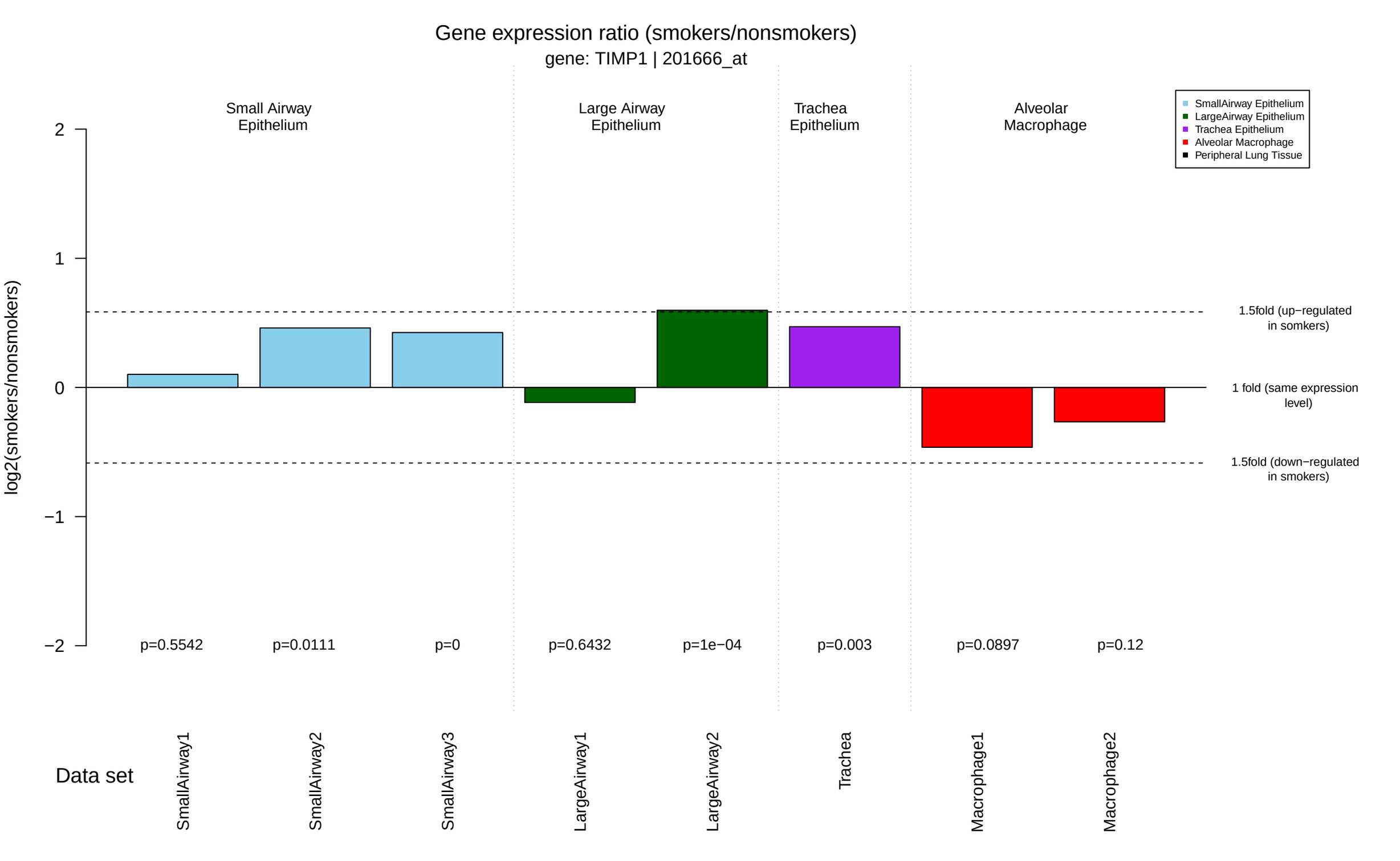
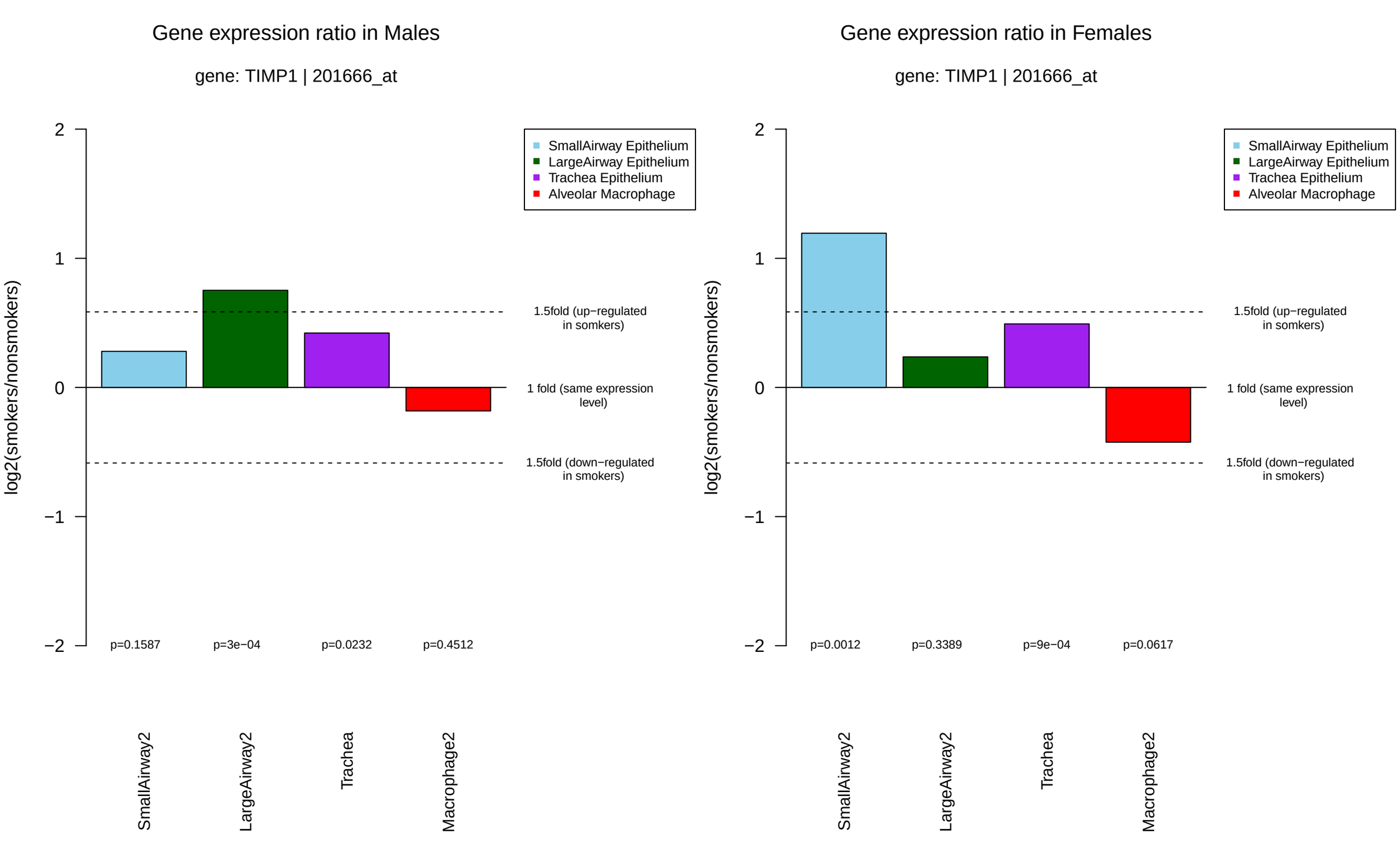
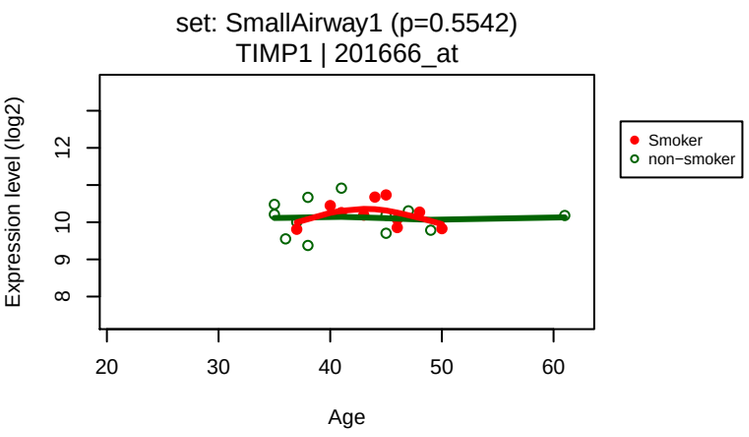
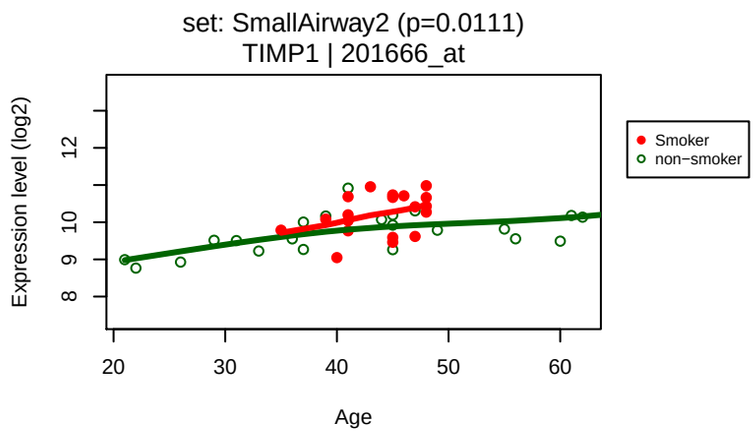
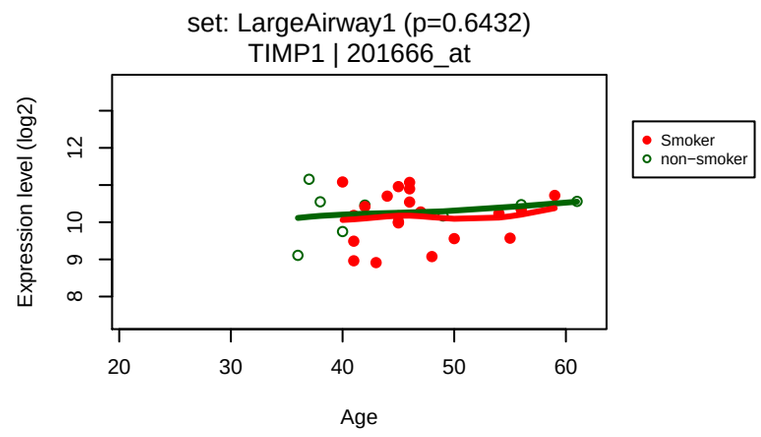

Gene expression of smokers and nonsmokers
gene: TIMP1 | 201666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5542
p=0.0111
p=0
p=0.6432
p=1e−04
p=0.003
p=0.0897
p=0.12
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

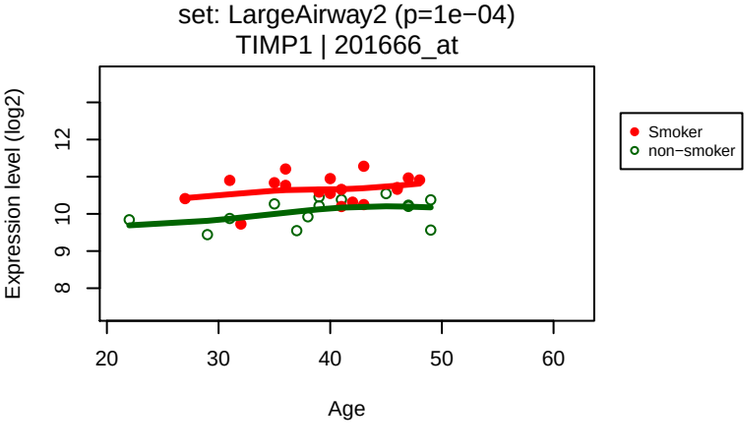

Gene expression ratio (smokers/nonsmokers)
gene: TIMP1 | 201666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5542
p=0.0111
p=0
p=0.6432
p=1e−04
p=0.003
p=0.0897
p=0.12
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

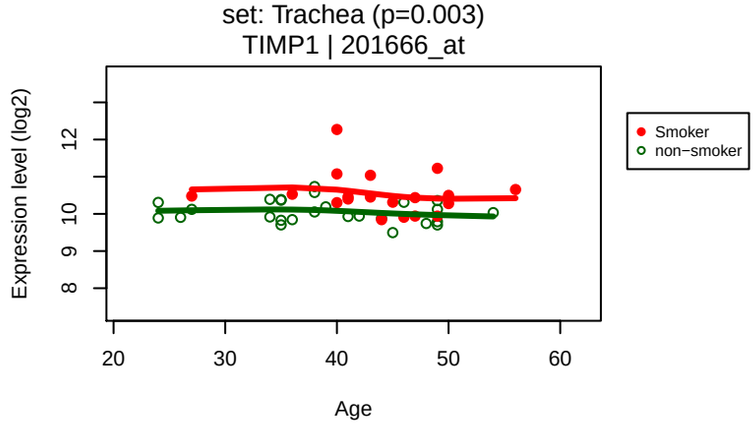

Gene expression ratio in Males
gene: TIMP1 | 201666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1587
p=3e−04
p=0.0232
p=0.4512
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP1 | 201666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.3389
p=9e−04
p=0.0617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

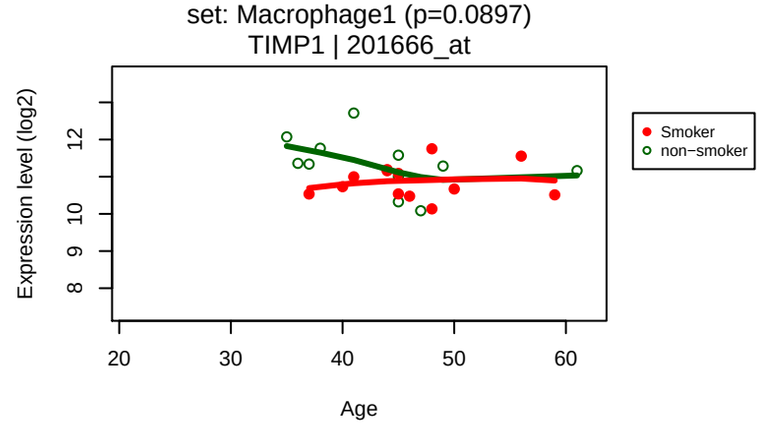

set: SmallAirway1 (p=0.5542)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker

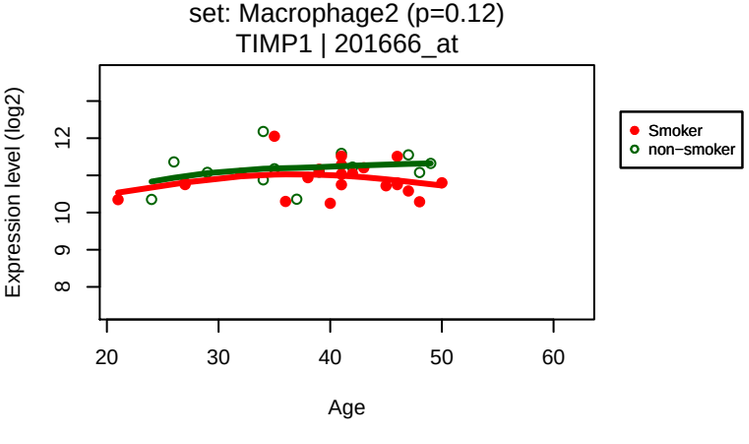

set: SmallAirway2 (p=0.0111)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker

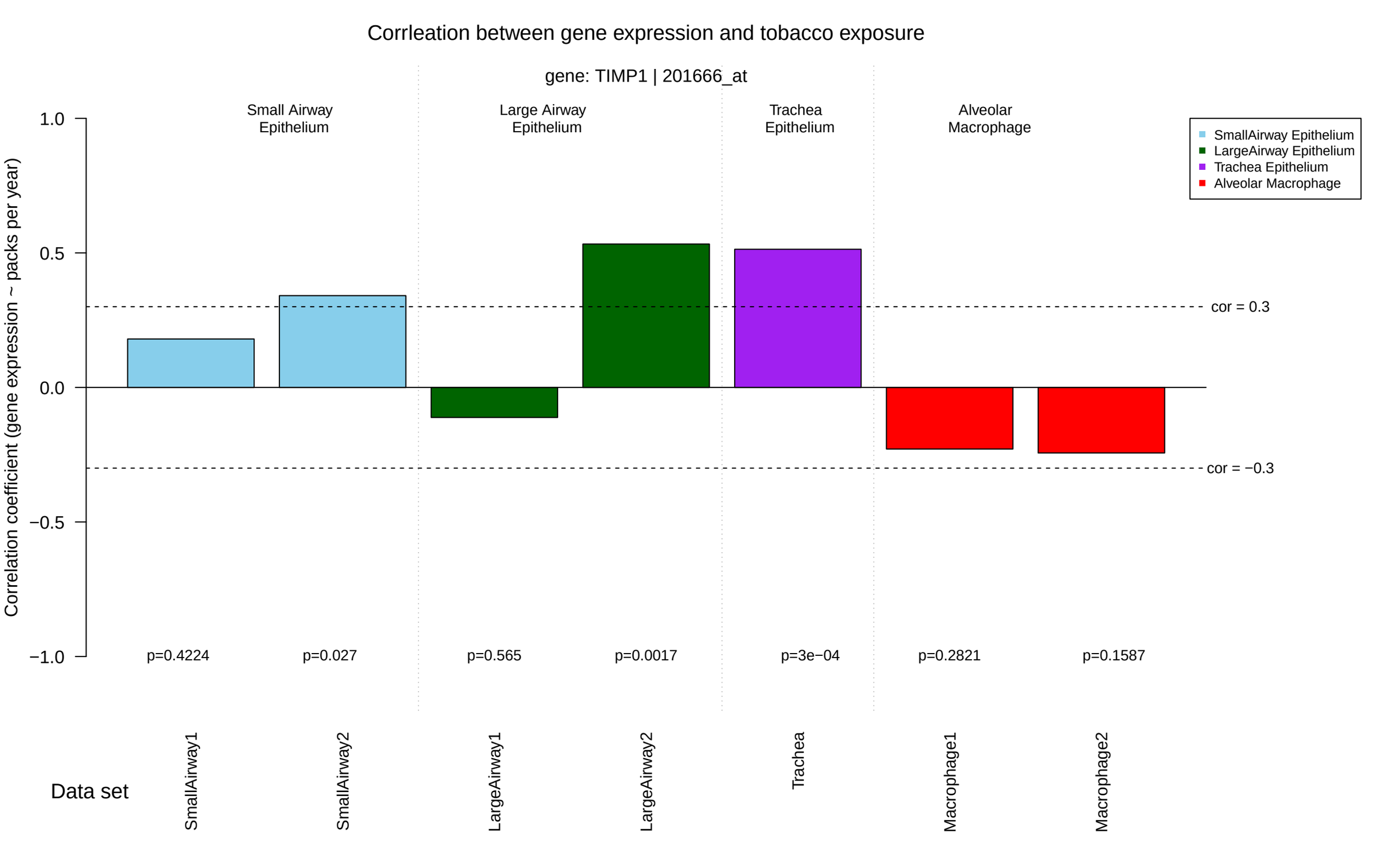

set: LargeAirway1 (p=0.6432)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=1e−04)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0897)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.12)
TIMP1 | 201666_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP1 | 201666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4224
p=0.027
p=0.565
p=0.0017
p=3e−04
p=0.2821
p=0.1587
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2